

RM21: Research Methodology			
Teaching Scheme: Theory Session: Total 45 Hours		Credit: 03	Examination Scheme: Internal (TH): 25 Marks External (TH): 50 Marks Total :75 Marks
Prerequisites: Fundamental knowledge of Statistics			
Course Objectives: - Understand fundamental research concepts and principles. - Develop skills to design and conduct research studies. - Learn to conduct thorough literature reviews and evaluate existing research. - Gain knowledge of various research designs and methodologies. - Acquire data collection and analysis skills using appropriate tools. - Understand and apply ethical considerations in research.			
Course Outcomes: On completion of the course, learners should be able to			
CO#	Cognitive Domain	Course Outcomes	
CO1	Understand	Understand the basic concepts, purposes, and significance of research methodology in academic and professional contexts.	
CO2	Apply	Apply various research designs and their appropriateness for different types of research questions and objectives	
CO3	Apply	Apply suitable data collection and sampling methods to gather reliable and valid data for research studies.	
CO4	Apply	Use appropriate statistical tools and techniques to demonstrate research data and interpret the results effectively.	
CO5	Apply	Apply skills in writing clear, coherent, and well-structured research reports that effectively communicate research findings.	
Unit No.	Contents		Weightage in %
1	Introduction to Research Methodology 1.1 Definition and Importance, Importance in academic and professional contexts 1.2 Objectives of Research-Exploration, Description, Explanation, Prediction, Application 1.3 Types of Research-Basic vs. applied research, Qualitative vs. quantitative research, Cross-sectional vs. longitudinal research 1.4 Research Process and Steps, Identifying the problem, Literature review, Research questions 1.5 Research in Computer Applications-Unique aspects of research in computer science, Common research methods in computer applications.		20
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*Mapping of Course Outcomes for Unit 1: CO1			

2	Research Design 2.1 Definition and Purpose, Importance of a well-structured design 2.2 Types of Research Designs: Exploratory design, Descriptive design, Experimental design, Quasi-experimental design 2.3 Components of Research Design: Objectives Hypotheses, Variables, Methods of data collection, Sampling design 2.4 Validity and Reliability: Internal validity, External validity, Construct validity, Reliability and consistency.	20	10
*Mapping of Course Outcomes for Unit 2: CO2			
3	Data Collection and Sampling Methods 3.1 Data Collection Methods: Primary Data Collection 3.2 Sampling Techniques: Principles of Sampling (Population vs. sample, Sampling frame), Probability Sampling Methods, Non-probability Sampling Methods, Determining Sample Size, Factors affecting sample size, Sample size calculations.	20	8
*Mapping of Course Outcomes for Unit3: CO3			
4	Data Analysis 4.1 Inferential Statistics: Hypothesis testing, Confidence intervals, Chi-square test, t-test, Analysis of variance (ANOVA) 4.2 Qualitative Data Analysis: Coding and Categorizing Data.	20	10
*Mapping of Course Outcomes for Unit 4: CO4			
5	Report Writing 5.1 Structure of a Research Report/Thesis Title Page, Abstract, Introduction, Literature Review, Methodology, Results, Discussion and Conclusion, References, Citation styles 5.2 Writing Style and Clarity-Academic writing standards, Avoiding plagiarism, Ensuring clarity and coherence, Visual Presentation of Data, Creating effective tables, Designing clear charts and graphs, Preparing for Oral Presentations, Structuring a presentation, Structuring a presentation, Use of visual aids, Effective communication skills, Ethics in Research.	20	10
*Mapping of Course Outcomes for Unit 5: CO5			
Learning Resources			

Text Books:

- Research Methodology: Methods and Techniques" by C.R. Kothari and Gaurav Garg
- Research Methodology: A Step-by-Step Guide for Beginners" by Ranjit Kumar
- Business Research Methods" by Donald R. Cooper and Pamela S. Schindler
- Research Methodology and Scientific Writing" by C. George Thomas
- Research Methodology: Concepts and Cases" by Deepak Chawla and Neena Sondhi

Reference Books:

- Research Methods for Business: A Skill Building Approach" by Uma Sekaran and Roger Bougie
- Qualitative Research Methods for the Social Sciences" by Bruce L. Berg and Howard Lune
- Statistics for Management" by Richard I. Levin, David S. Rubin, Masood H. Siddiqui, and Sanjay Rastogi
- Case Study Research: Design and Methods" by Robert K. Yin
- The Craft of Research" by Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams

Recommended Learning Material

Online Courses

- Coursera: "Research Methods" by University of London
- A comprehensive course covering the fundamentals of research methods.
- edX: "Introduction to Research Methods" by University of London
- Focuses on essential research methods and techniques.
- Udemy: "Research Methods and Statistics: An Introduction"
- Provides an introduction to research methods and basic statistical concepts.
- Khan Academy: "Statistics and Probability"
- Offers free tutorials on statistical methods relevant to research.
- MIT OpenCourseWare: "Research Methods in Management"
- A free course providing an overview of research methods in management.

Software Tools

- SPSS (Statistical Package for the Social Sciences)
- R and RStudio
- Microsoft Excel
- Software Tools-SPSS, R, Python (Pandas, NumPy)

Recommended Certification

- Research Methodology, Link: <https://nptel.ac.in/courses/109/105/109105115/>
- Qualitative Research Methods and Research Writing, Instructor: Prof. Aradhna Malik, IIT Kharagpur, Link: <https://nptel.ac.in/courses/109/105/109105115/>
- Introduction to Research, Link: <https://nptel.ac.in/courses/121/106/121106007/>
- Research Writing, Instructor: Prof. A. Arunachalam, IIT Kharagpur, Link: <https://nptel.ac.in/courses/109/105/109105115/>
- Advanced Statistical Methods in Data Science, Instructor: Prof. Shalabh, IIT Kanpur Link: <https://nptel.ac.in/courses/111/104/111104071/>